

On the bottleneck partial terminal Steiner tree problem *

Yen Hung Chen

Department of Computer Science, Taipei Municipal University of Education,
Taipei 10042, Taiwan, R.O.C.
{yhchen}@tmue.edu.tw

Abstract

Given a complete graph $G = (V, E)$, a positive length function on edges, and two subsets R of V and R' of R , the partial terminal Steiner tree problem is defined to be an acyclic subgraph of G spanning all vertices in R such that each vertex of R' must belong to a leaf of the subgraph. The bottleneck partial terminal Steiner tree problem is to find a partial terminal Steiner tree problem T for R and R' in G such that the length of the largest edge in T is minimized. In this paper, we present the first known exact algorithm of $O(|E| \log |E|)$ time to solve the bottleneck partial terminal Steiner tree problem.

Keywords: Steiner tree, partial terminal Steiner tree problem, bottleneck partial terminal Steiner tree problem, (sensor) network facility allocation, engineering change orders (ECO) in VLSI design

1 Introduction

The Steiner tree problem (STP) is one of the fundamental problems in network design [7, 12, 11, 22]. Given an arbitrary graph $G = (V, E)$, a subset $R \subset V$ of vertices, and a positive length function on edges, a Steiner tree is used to find a connected and acyclic subgraph of G that spans all vertices in R . The given vertices R are usually referred to as *terminals* and other vertices $V \setminus R$ as *Steiner vertices*. The length of a Steiner tree is defined to be the sum of the lengths of all its edges. The STP is concerned with the determination of a Steiner tree with minimum length in G [7, 12, 11, 22]. The STP was shown to be NP-complete [15] and MAX SNP-hard [2]. Hence, some polynomial-time approximation algorithms with constant ratios had been

proposed [1, 3, 4, 17, 24, 30, 31, 32, 33] instead of exact algorithms for the STP. The STP has many important applications in VLSI design, network communication, computational biology and so on [5, 7, 12, 11, 16, 22, 23, 25].

Motivated by the (sensor) network facility allocation and engineering change orders (ECO) in VLSI design, Hsieh and Gao [19] presented a variant of the STP, called the *partial terminal Steiner tree problem* (PTSTP). Given a complete graph $G = (V, E)$, a positive length function on edges, and two subsets $R \subset V$ and $R' \subseteq R$, a Steiner tree for R in G is a partial terminal Steiner tree if all terminals in R' must be leaves of this Steiner tree. The *partial terminal Steiner tree problem* (PTSTP) is concerned with finding a partial terminal Steiner tree for R and R' in G whose total edge length is minimized [19, 18]. A special case of the PTSTP is called as the *terminal Steiner tree problem* (TSTP). The TSTP is assumed that $R' = R$ and the purpose is to find a minimum total edge length Steiner tree for R in G such that all vertices of R must be leaves of T [3, 12, 22, 27, 28]. Since the TSTP had been shown to be NP-complete and MAX SNP-hard [27, 28], Chen, Lu and Tang [6], Fuchs [14], Drake and Hougary [10], proposed 2ρ -approximation algorithms for the TSTP on metric graphs (i.e., a complete graph and the lengths of edges satisfy the triangle inequality), independently, where ρ is the best-known performance ratio for the STP [4]. Then Martinez, Pinab, and Soares [29] designed a $(2\rho - (\frac{\rho}{3\rho-2}))$ -approximation algorithm which has the current best performance ratio for the TSTP on metric graphs. Since the TSTP is a special case of the PTSTP, the PTSTP is also NP-complete and MAX SNP-hard. Hsieh and Pi [18] presented a $(2\rho - (\frac{\rho}{3\rho-2}))$ -approximation algorithm for the PTSTP on metric graphs.

Although the PTSTP is defined by a min-sum objective function, some applications of network and VLSI routing are considered in the bottleneck

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(min-max) objective function [6, 7, 12, 11, 22]. Hence, we propose a variant of the PTSTP with bottleneck objective function, called as the bottleneck partial terminal Steiner tree problem. Given a complete graph $G = (V, E)$, a positive length function on edges, and two subsets $R \subset V$ and $R' \subseteq R$, the bottleneck edge of a Steiner tree is an edge with the largest length in the Steiner tree. The *bottleneck partial terminal Steiner tree problem* (BPTSTP) is concerned with the determination of a partial terminal Steiner tree for R and R' in G with minimum length of the bottleneck edge. For other related problems of the BPTSTP, the bottleneck Steiner tree problem (BSTP) is to find a Steiner tree T in G with minimum length of the bottleneck edge [8, 13]. However, the bottleneck Steiner tree problem can be solved exactly in polynomial time [13]. Chen, Lu and Tang [6] also defined a bottleneck version of the TSTP, called as the bottleneck terminal Steiner tree problem (BTSTP). BTSTP is to find a Steiner tree T for R in G with minimum length of the bottleneck edge such that all vertices of R must be leaves of T . The BTSTP can also be solved exactly in polynomial time [6]. A contrary problem of PTSTP is called the *selected-internal Steiner tree problem* (SISTP) which is to find a minimum length Steiner tree T for R and R' in G such that all vertices of R' must be internal vertices of T [20]. Then Hsieh and Yang [20] showed that the SISTP is NP-complete and MAX SNP-hard. They also presented a 2ρ -approximation algorithm for the SISTP on *metric graphs*, where ρ is the best-known performance ratio for the STP (currently $\rho = \ln 4 + \epsilon \approx 1.39$ [4]). Li et al. [26] designed a $(\rho + 1)$ -approximation algorithm which has the current best performance ratio for the SISTP on metric graphs. A similar problem of the SISTP is called as the *internal Steiner tree problem* (ISTP) is assumed that $R' = R$ and the purpose is to find a selected-internal Steiner tree for R in G with minimum total edge length. Huang et al. [21] gave a $(2\rho + 1)$ -approximation algorithm for the ISTP on metric graphs. In this paper, we give the first known exact algorithm of $O(|E| \log |E|)$ time to solve the BPTSTP.

2 An Exact Algorithm for the BPTSTP

BPTSTP (Bottleneck partial terminal Steiner Tree Problem)

Instance: A complete graph $G = (V, E)$ with a positive length function $\ell : E \rightarrow R^+$ on the edges, and two subsets $R \subset V$ and $R' \subseteq R$.

Problem: Find a partial terminal Steiner tree for R and R' in G such that the length of the bottleneck edge is minimized.

In the section, we will present an exact algorithm for solving the above BPTSTP in $O(|E| \log |E|)$ time. By definition, any partial terminal Steiner tree T for R and R' in $G = (V, E)$ contains no edge in $E_R = \{(u, v) | u, v \in R', u \neq v\}$. Hence, throughout the rest of this paper, we assume that G contains no edge in E_R (i.e., $E \cap E_R = \emptyset$). Then we call the terminal vertices in R' are usually referred to as *partial terminal (PT) vertices* and other vertices $V \setminus R'$ as *non partial terminal (NPT) vertices*. We also call an edge in E as *non partial terminal (NPT) edge* if both its end-vertices are NPT vertices; otherwise, as *partial terminal (PT) edge*. Without loss of generality, we use $\{e_1, e_2, \dots, e_\beta\}$ to denote the set of all NPT edges in E and assume that their lengths are different (i.e., $\ell(e_1) < \ell(e_2) < \dots < \ell(e_\beta)$). For any arbitrary tree T , let $len_b(T)$ denote the length of the bottleneck edge.

In the following, we describe our algorithm to find an optimal bottleneck partial terminal Steiner tree for R and R' in G . In the beginning, we construct a star T_u for each NPT vertex $u \in V \setminus R'$ by letting u as the center of T_u and all PT vertices in R' as the leaves of T_u . Then we let $\mathcal{C}$ denote the collection of all such stars (i.e., $\mathcal{C} = \{T_u | u \in V \setminus R'\}$). If there exists a feasible solution in $\mathcal{C}$ for the BPTSTP, we let $T^*(\mathcal{C})$ denote such the partial terminal Steiner tree. Otherwise, $T^*(\mathcal{C})$ does not exist. It is not hard to see that if $T^*(\mathcal{C})$ exists and $len_b(T^*(\mathcal{C})) \leq \ell(e_1)$, then $T^*(\mathcal{C})$ is the optimal solution for the BPTSTP, since we have $len_b(T^*(\mathcal{C})) \leq len_b(T)$ for any partial terminal Steiner tree T in G containing at least one NPT edge. On the other hand, if $T^*(\mathcal{C})$ does not exist or $len_b(T^*(\mathcal{C})) > \ell(e_1)$, then we perform the following Algorithm 1 to merge some trees in $\mathcal{C}$ into larger trees containing more NPT vertices, and then we can show later that the partial terminal Steiner tree in the resulting $\mathcal{C}$ is an optimal bottleneck partial terminal Steiner tree in G . For each terminal vertex $r \in R'$, we use $e_T(r, T)$ to denote the PT edge in a partial terminal Steiner tree T incident with r .

Algorithm 1: Repeatedly merge the trees in $\mathcal{C}$

1. Let $e_i = e_1$.
2. Repeat the following steps until the condition (1) $T^*(\mathcal{C})$ is a partial terminal Steiner tree and $\text{len}_b(T^*(\mathcal{C})) \leq \ell(e_i)$ or (2) $|\mathcal{C}| = 1$ holds.
 - 2.1. Let $v_1(e_i)$ and $v_2(e_i)$ be the end-vertices of e_i . Find the trees in $\mathcal{C}$, say T_{i_1} and T_{i_2} respectively, containing $v_1(e_i)$ and $v_2(e_i)$.
 - 2.2. If $T_{i_1} = T_{i_2}$ then let $T_i = T_{i_1}$ else
 - 2.2.1. Connect T_{i_1} with T_{i_2} by e_i and let T_i be the resulting tree.
 - 2.2.2. /* Make sure each PT vertex $r \in R'$ is adjacent to the nearest NPT vertex v in T_i (i.e., $\ell(r, v) = \min\{\ell(r, v') \mid v' \text{ is a NPT vertex in } T_i\}$). */
For each $r \in R'$, if $\ell(e_T(r, T_{i_1})) \leq \ell(e_T(r, T_{i_2}))$ then delete $e_T(r, T_{i_2})$ in T_i else delete $e_T(r, T_{i_1})$ in T_i .
 - 2.2.3. /* Check whether there exists a feasible solution in $\mathcal{C}$ for the BPTSTP */
if T_i is a partial terminal Steiner tree then let $T^*(\mathcal{C}) = T_i$.
- 2.3. Let $i = i + 1$.

If there exists the first feasible solution (a partial terminal Steiner tree) $T^*(\mathcal{C})$ of the BPTSTP for running Algorithm 1, it is not hard to see that after running each iteration of Algorithm 1, $T^*(\mathcal{C})$ may have a better bottleneck edge with smaller length. After running Algorithm 1, it exists a feasible solution for the BPTSTP and let $T^*(\mathcal{C})$ be the output of our algorithm for the BPTSTP in the resulting $\mathcal{C}$. For clarification, we describe our algorithm for the BPTSTP as follows.

Algorithm Exact-BPTSTP

Input: A complete graph $G = (V, E)$ and two set $R \subset V$ and $R' \subseteq R$, where we assume that G contains no edge in E_R .

Output: An optimal bottleneck partial terminal Steiner tree T_{OPT} for R and R' in G .

1. For each NPT vertex $u \in V \setminus R'$, construct a star T_u by letting u as its center and all PT vertices in R' as its leaves, and let $\mathcal{C} = \{T_u \mid u \in V \setminus R'\}$.
2. If there exists a feasible solution in $\mathcal{C}$ for the BPTSTP, let $T^*(\mathcal{C})$ denote such the partial terminal Steiner tree.

3. Sort all NPT edges of G into an increasing order, say $e_1, e_2, \dots, e_\beta$, with $\ell(e_1) < \ell(e_2) < \dots < \ell(e_\beta)$.

4 Perform Algorithm 1.

- 5 Repeatedly remove all Steiner vertices of $T^*(\mathcal{C})$ whose degrees are 1.

6. Let $T_{OPT} = T^*(\mathcal{C})$.

Theorem 1: Algorithm Exact-BPTSTP is an $O(|E| \log |E|)$ time algorithm for solving the BPTSTP.

Proof: We first analyze the time-complexity of Algorithm Exact-BPTSTP as follows. It is not hard to see that its time-complexity is dominated by the steps 3. Clearly, step 3 can be done in $O(|E| \log |E|)$ time by heap sorting algorithm [9]. For step 4 of executing Algorithm 1, it can be implemented by the disjoint-set operations (i.e., MAKE-SET, UNION and FIND-SET operations), since for each tree of $\mathcal{C}$, we can represent it by using the set of its NPT vertices and clearly, any two such sets are disjoint. Initially, we make $|V \setminus R'|$ singleton sets with each consisting of a NPT vertex u and representing an initial T_u in $\mathcal{C}$. After that, in each iteration of Algorithm 1, we can use two FIND-SET operations to identify T_{i_1} and T_{i_2} , and then use a UNION operation to merge them into T_i . Totally, we have $|V \setminus R'|$ MAKE-SET operations, at most $2 \cdot |E|$ FIND-SET operations, and at most $|V \setminus R'| - 1$ UNION operations. Note that the cost of m disjoint-set operations on n elements takes $O(m \cdot \alpha(n))$, where $\alpha(n)$ is an inverse Ackermann's function of n which is almost a constant for all practical purposes (i.e., $\alpha(n) \leq 4$ for all $n \leq 10^{80}$) [9]. In other words, above operations can be implemented in $O(|V \setminus R'| + 2 \cdot |E| + |V \setminus R'| - 1) = O(|E|)$ time. Clearly, the total cost of step 2.2.2 of Algorithm 1 is at most $O(|V \setminus R'| \times |R'|) = O(|V|^2)$ time. The cost of step 2.2.3 of Algorithm 1 (check whether there exists a partial terminal Steiner tree in $\mathcal{C}$) takes $O(|V|)$ time. Hence, the total time-complexity of Algorithm 1 is $O(|E|)$ time. As mentioned above, the time-complexity is $O(|E| \log |E|)$.

Next, we prove the correctness of Algorithm Exact-BPTSTP by showing that for any partial terminal Steiner tree T for R and R' in G , we have $\text{len}_b(T_{OPT}) \leq \text{len}_b(T)$, where T_{OPT} is the output of our algorithm. Let $V_{NP}(T)$ be the set of the NPT vertices in T . Suppose that $|V_{NP}(T)| = 1$. Then T must be a partial terminal Steiner tree in the initial $\mathcal{C}$ of Algorithm

Exact-BPTSTP. Clearly, if there exists a partial terminal Steiner tree in $\mathcal{C}$, after performing each iteration of Algorithm 1, $\text{len}_b(T^*(\mathcal{C}))$ may be improved. Hence, $\text{len}_b(T_{\text{opt}}) \leq \text{len}_b(T)$. Suppose that $|V_{NP}(T)| \geq 2$. Let e_γ be the edge with the largest length in NPT edges of T . Assume that Algorithm Exact-BPTSTP outputs its result T_{OPT} after its Algorithm 1 considers NPT edge e_k , where $1 \leq k \leq \beta$. Then we consider the following two cases. Case 1: $k \leq \gamma$. In this case, we have either $\text{len}_b(T_{OPT}) \leq \ell(e_k)$ or the resulting $\mathcal{C}$ has size 1. For the former case, we have $\text{len}_b(T_{OPT}) \leq \text{len}_b(T)$ since $\ell(e_k) \leq \ell(e_\gamma) \leq \text{len}_b(T)$. For the latter case, it is clear that e_{k-1} is the largest NPT edge of T_{OPT} , whose length is less than $\ell(e_\gamma)$. Moreover, we have $V_{NP}(T) \subseteq V_{NP}(T_{OPT})$ since $V_{NP}(T_{OPT}) = V \setminus R'$, which implies that for each terminal $r \in R'$, its corresponding PT edge in T_{OPT} has length smaller than or equal to that in T . The reason is that our algorithm always connects r to the nearest NPT vertex in T_{OPT} , which contains all NPT vertices. As discussed above, we have $\text{len}_b(T_{OPT}) \leq \text{len}_b(T)$. Case 2: $k > \gamma$. For each NPT edge of T being considered by our algorithm, it is not hard to see that there is a tree in $\mathcal{C}$ containing its two end-vertices. Since the subgraph of T induced by $V_{NP}(T)$ is connected, T_γ in $\mathcal{C}$ contains all vertices in $V_{NP}(T)$ after e_γ is considered (i.e., $V_{NP}(T) \subseteq V_{NP}(T_\gamma)$). Hence, T_γ is a partial terminal Steiner tree in $\mathcal{C}$. Then for each terminal $r \in R'$, its corresponding PT edge in T_γ has length smaller than or equal to that in T . In addition, the largest NPT edge of T_γ has length less than or equal to $\ell(e_\gamma)$. Hence, we have $\text{len}_b(T_\gamma) \leq \text{len}_b(T)$. Since T_γ is a partial terminal Steiner tree, we have $\text{len}_b(T_{OPT}) \leq \text{len}_b(T_\gamma)$ and hence we have $\text{len}_b(T_{OPT}) \leq \text{len}_b(T)$. $\square$

3 Conclusion

In this paper, we presented the first known exact algorithm of $O(|E| \log |E|)$ time for solving the BPTSTP. For future research, improving the time complexity of exact algorithm for the BPTSTP is an interesting direction.

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